STAT140 XP
LAPD Crime Analysis

Group 1

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Team 1: 2020-2024 LA Crime Data Analysis

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Abstract

We analyzed the LAPD public crimes dataset to explore patterns in crime reports, victim demographics, reporting behavior, and temporal activity in Los Angeles County. The study examines relationships between time of day and crime type, victim demographics and crime frequency, and the time between when a crime occurs and when it is reported. Crimes were categorized into time periods (Midday, Evening, Night, and Late Night), and chi-squared tests were used to evaluate associations between variables. Results show that crime patterns vary throughout the day with significant associations between crime type and time period. Victim age was not significantly associated with crime frequency, while sex showed a significant relationship with crime type. Additionally, a significant association was found between crime type and the time between occurrence and reporting. These findings help provide insight into crime trends in Los Angeles County and may support efforts to better understand and address public safety concerns.

Motivation

Los Angeles County experiences persistent crime risk that varies by time and victim characteristics. Understanding these patterns helps reveal when incidents are most likely to occur. This analysis aims to communicate these trends and support future data-driven approaches to improving public safety.

Data

Source: LAPD Crime Data (LAPD CRIME DATA.csv)
Time window: 2020–2023 (2024+ excluded due to incomplete reporting patterns)
Unit: One incident report (DR_NO)
Key fields: DATE.OCC, TIME.OCC, AREA.NAME, Crm.Cd.Desc, Vict.Age, Vict.Sex, Weapon.Used.Cd, Premis.Desc, DATE.RPTD, TIME.RPTD
Cleaning choices: Dropped Crm.Cd.2–4 and Cross.Street due to sparsity and low relevance for our main questions.

Research Questions

- Is these a relationship between the time of day and the type of crime committed in Los Angeles County?
- Is there a relationship between victim demographics (age and sex) and the frequency or type of crime?
- Is there an association between the type of crime and the time between when the crime occurred and when it was reported?

Literature Review

Los Angeles has been known to be a crime-ridden county, and many individuals have studied the rise and fall of crime throughout time. Jim Newton from CalMatters discusses how Los Angeles' crime rate has dropped in recent years, yet still persists while LAPD's numbers decrease year by year. He mentions that property and personal theft crimes are the highest reported incidents, and that homicide and rape are among those that have decreased. Thus, our study further expands on the types of crimes committed over time, to gain more insight on the L.A. county's potential dangers. There have also been examinations on how crime in Los Angeles is related to zoning and land use, with James Anderson of University of Pennsylvania Law Review positing that areas with denser commercial establishments create more attractive opportunities for crime.

This study fits similar lines of the proposal of our city by looking into similar variables such as the measure of crime type or measuring the frequency of the data values over the past couple of years which is feasible due to the data we have in the past 5 years. Additionally the articles have similar goals of trying to learn lesson from the data in hopes to help keep Los Angeles County safe and have positive progress moving forward in hopes for a safer community

Hypotheses (Testable)

Time and Violence

- **H0:** The share of violent crimes does not increase later in the day.
- **H1:** Violent crimes are more common in evening/night hours (18:00–23:59).

Victim Demographics

- **H0:** Victim demographics are not associated with crime type.
- **H1:** Certain crime types show differences by sex or age distribution.

Reporting Time

- **H0:** There is no association between crime type and the time between when a crime occurs and when it is reported.
- **H1:** There is an association between crime type and the time between when a crime occurs and when it is reported.

Results

What the Data Shows (Summary Statistics)

- Crime is distributed across multiple LAPD divisions with Cnetral, 77th Street, and Southwest reporting the most incidents
- Vehicle theft is the most common crime; property crime dominates overall
- Victims are primarily young and middle-aged adults (20 - 40 years)
- Temporal trends: crimes are lowest in early morning (4 - 6 AM) and peak midday/evening
- Most crimes are reported quickly, usually the same day
- Gender differences exist, but do not strongly determine victimization patterns

Methods

Exploratory Data Analysis - Analyzed LAPD crime dataset (2020-2024) with 877,327 incidents - Focused on primary crime; removed secondary/tertiary crimes and incomplete 2024 - Examined temporal patterns such as hour of day and reporting delays - Investigated victim demographics, weapon involvement, and crime tpe composition - Visualized patterns using line graphs, box plots, and mosaic plots

Planned Follow-Up Analysis - Apply ANOVA and regression to test relationships between time, area, and crime type - Explore predictive modeling for reporting delays and crime occurrence - Examine correlations between crime and neighborhoods characteristics, land use, or demographics - Analyze weapon-related patterns in relation to crime type and time

Key Findings

- Property crimes dominate reported evidence. Vehicle theft, burglary, and identity theft appear the most frequently across nearly all LAPD divisions
- Crime mainly affects working-age adults (20-40)
- Crime follows daily activity patterns and is reported quickly. Incidents peak during midday and evening hours, and most crimes are reported the same day or shortly after occurring

Limitations

- The dataset only includes reported crimes. Unreported incidents are not captures, which may underestimate the true level of crime
- Some data had to be removed during cleaning. Incomplete 2024 records and variable with many missing values were excluded from the analysis.
- Our analysis is descriptive rather than causal. It identifies patterns in the data, but does not determine why those patterns occur.

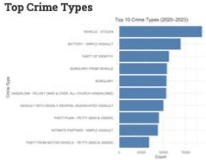
Future Work:

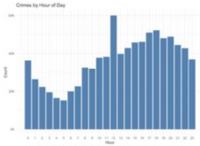
- Apply predictive modeling to examine factor influencing crime occurrence and reporting delays
- Explore relationships between crime and neighborhood characteristics, economic conditions, and land use
- Investigate weapon-related patterns and correlations with crime type or time of day

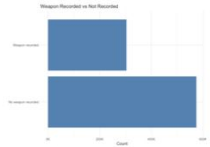
Results and Visuals

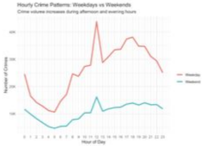
Overall, the statistical tests provide evidence that crime patterns in Los Angeles are shaped by demographic characteristics, geographic variation, and temporal activity patterns rather than occurring randomly across the city. These findings highlight the importance of considering multiple factors when analyzing urban crime data.

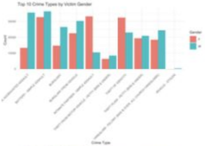
Top Crime Types

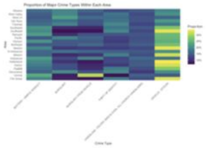












Motivation/Data

Los Angeles County experiences persistent crime risk that varies by time and victim characteristics. Understanding these patterns helps reveal when incidents are most likely to occur. This analysis aims to communicate these trends and support future data-driven approaches to improving public safety.

Source: LAPD Crime Data (LAPD CRIME DATA.csv)

Time window: 2020-2023 (2024+ excluded due to incomplete reporting patterns)

Unit: One incident report (DR_NO)

Key Xelds: DATE.OCC, TIME.OCC, AREA.NAME, Crm.Cd.Desc, Vict.Age, Vict.Sex, Weapon.Used.Cd, Premis.Desc, DATE.RPTD, TIME.RPTD

Cleaning choices: Dropped Crm.Cd.2-4 and Cross.Street due to sparsity and low relevance for our main questions.

Research Questions

1. Is there a relationship between the time of day and the type of crime committed in Los Angeles County?

2. Is there a relationship between victim demographics (age and sex) and the frequency or type of crime?

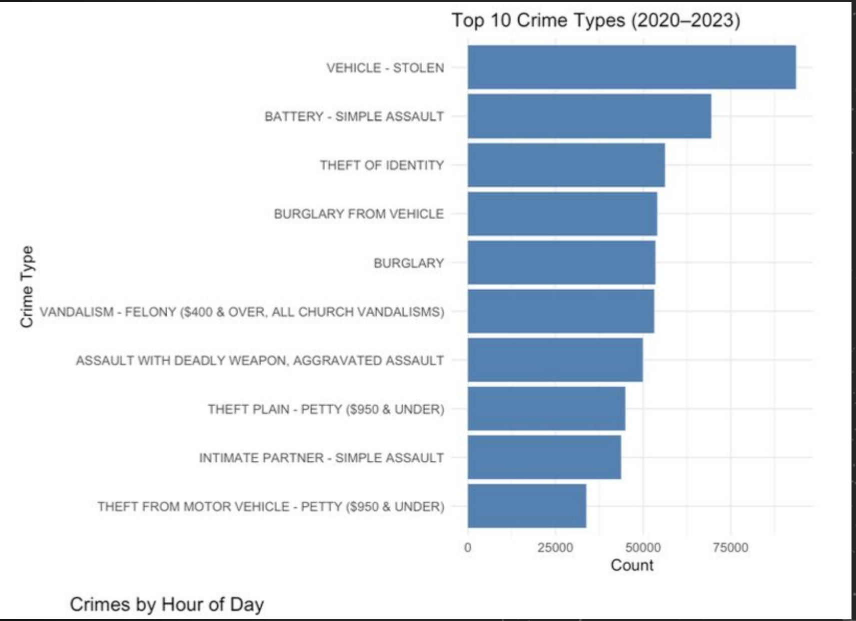
3. Is there an association between the time of crime and the time between when the crime occurred and when it was reported?

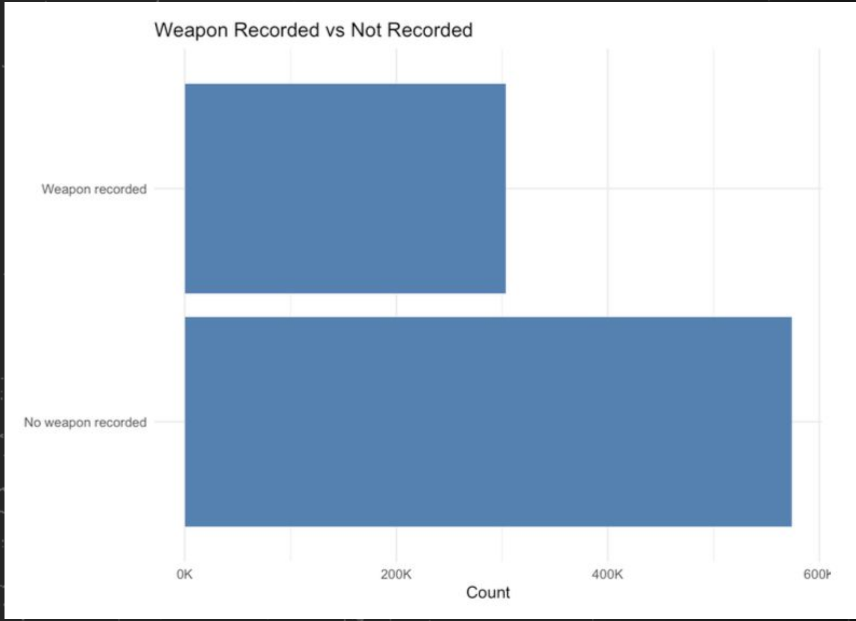
Literature Review

- Jim Newton, CalMatters: Los Angeles' crime rate has dropped in recent years, yet still persists while LAPD's numbers decrease year by year
  - property and personal theft crimes are the highest reported incidents
  - homicide and rape are among those that have decreased
- James Anderson, University of Pennsylvania Law Review: zoning and land use
  - areas with denser commercial establishments create more attractive opportunities for crime

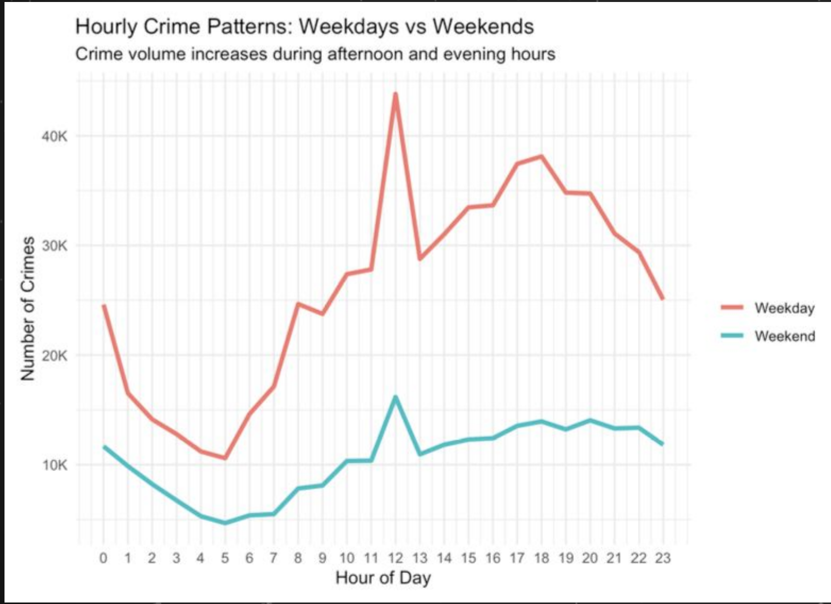
This study builds on these findings by examining crime types and their frequency over the past 3 years, with the shared goal of better understanding LA County's risks and fostering a safer community.

EDA Visuals

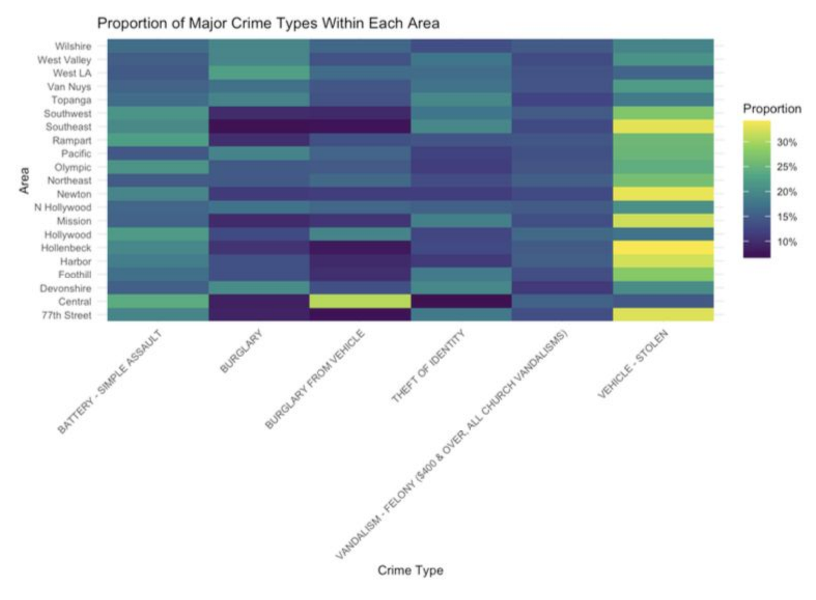

Top 10 Crime Types (2020–2023)
Crime Type
VEHICLE - STOLEN
BATTERY - SIMPLE ASSAULT
THEFT OF IDENTITY
BURGLARY FROM VEHICLE
BURGLARY
VANDALISM - FELONY ($400 & OVER, ALL CHURCH VANDALISMS)
ASSAULT WITH DEADLY WEAPON, AGGRAVATED ASSAULT
THEFT PLAIN - PETTY ($950 & UNDER)
INTIMATE PARTNER - SIMPLE ASSAULT
THEFT FROM MOTOR VEHICLE - PETTY ($950 & UNDER)
0
25000
50000
75000
Count
Crimes by Hour of Day


Weapon Recorded vs Not Recorded
Weapon recorded
No weapon recorded
0K
200K
400K
600K
Count

EDA Visuals



Hourly Crime Patterns: Weekdays vs Weekends
Crime volume increases during afternoon and evening hours
Number of Crimes
40K
30K
20K
10K
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Hour of Day
Weekday
Weekend



Proportion of Major Crime Types Within Each Area
Area
Wilshire
West Valley
West LA
Van Nuys
Topanga
Southwest
Southeast
Rampart
Pacific
Olympic
Northeast
Newton
N Hollywood
Mission
Hollywood
Hollenbeck
Harbor
Foothill
Devonshire
Central
77th Street
BATTERY - SIMPLE ASSAULT
BURGLARY
BURGLARY FROM VEHICLE
THEFT OF IDENTITY
VANDALISM - FELONY ($400 & OVER, ALL CHURCH VANDALISMS)
VEHICLE - STOLEN
Crime Type
Proportion
30%
25%
20%
15%
10%

Planned Follow Up Analysis

- Planned Follow-Up Analysis
    - Apply ANOVA and regression to test relationships between time, area, and crime type
    - Explore predictive modeling for reporting delays and crime occurrence
    - Examine correlations between crime and neighborhoods characteristics, land use, or demographics
    - Analyze weapon-related patterns in relation to crime type and time

Hypotheses (Testable)

Time and Violence:

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Victim Demographics

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Reporting Time

- H0: There is no association between crime type and the time between when a crime occurs and when it is reported.
- H1: There is an association between crime type and the time between when a crime occurs and when it is reported.

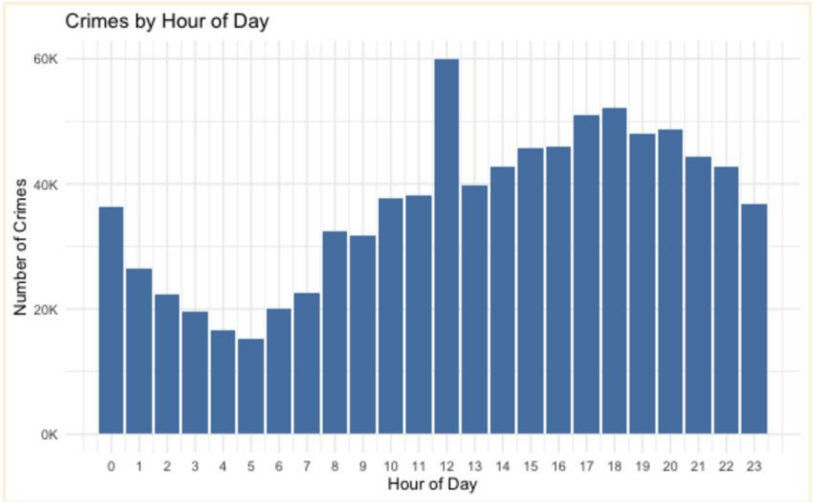
H1: Time and Violence

Midday (11:00 - 14:00) vs Evening (18:00 - 22:00)

- Compared the proportion of violent crimes between:
- Chi-square test to determine if the proportions differ
- Statistically significant therefore violent crimes make up a significantly larger proportion during evening hours

Night (18:00 - 21:00) vs Late Night (22:00 - 03:00)

- Compared the proportion of violent crimes between:
- Chi-square test to determine if the proportions differ
- Statistically significant therefore violent crimes decrease slightly later at night



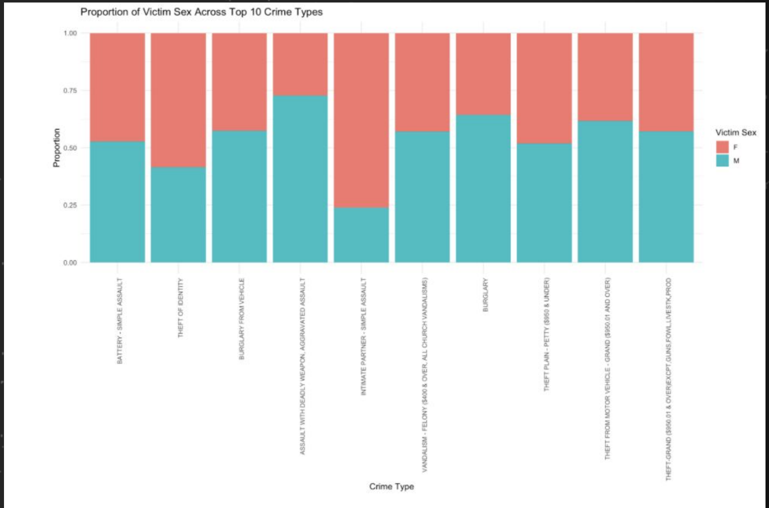
H2: Victim Demographic

Age and Crime Frequency

- Cleaned age data to have only above 0
- Chi-square test to determine if the proportions differ
- Statistically significant therefore negative relationship between age and crime frequency

Sex and Crime Type

- Claned to only have female and male based on distribution
- Chi square test to determine if the proportions differ
- Statistically significant therefore genders are prone to some specific crime types



H3: Time Occured and Time Reported

Time and Time until Report

- Used same time categorization from earlier
- Used a chi-square test to determine if the proportions differ
- Statistically significant therefore there are specific reporting delays which are associated with the time of the day that the crime originally occurred

	1-3 days	More than 3 days	Same day	Within 1 hour
Afternoon	0.103	0.240	0.182	0.475
Late Night	0.068	0.160	0.092	0.681
Morning	0.083	0.229	0.120	0.569
Night	0.116	0.150	0.377	0.357

Key Findings

Property crimes dominate reported evidence

Vehicle theft, burglary, and identity theft appear the most frequently across nearly all LAPD divisions

Crime mainly affects working-age adults (20-40) and decreases there after

Crime follows daily activity patterns and is typically relatively reported quickly

Incidents peak during midday and evening hours

Limitations

The dataset only includes reported crimes, which may underestimate the true level of crime.

Some data had to be removed during cleaning. Incomplete 2024 records and variable with many missing values were excluded from the analysis.

Our analysis is descriptive rather than causal. It identifies patterns in the data, but does not determine why those patterns occur.

Future Work

Apply predictive modeling to examine factor influencing crime occurrence and reporting delays

Explore relationships between crime and neighborhood characteristics, economic conditions, and land use

Investigate weapon-related patterns and correlations with crime type or time of day

Thank You

Any Questions?